Parsing Façade with Rank-One Approximation

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Abstract

The binary split grammar is powerful to parse façade in a broad range of types, whose structure is characterized by repetitive patterns with different layouts. We notice that, as far as two labels are concerned, BSG parsing is equivalent to approximating a façade by a matrix with multiple rank-one patterns. Then, we propose an efficient algorithm to decompose an arbitrary matrix into a rank-one matrix and a residual matrix, whose magnitude is small in the sense of l^0-norm. Next, we develop a block-wise partition method to parse a more general façade. Our method leverages on the recent breakthroughs in convex optimization that can effectively decompose a matrix into a low-rank and sparse matrix pair. The rank-one block-wise parsing not only leads to the detection of repetitive patterns, but also gives an accurate façade segmentation. Experiments on intensive façade data sets have demonstrated that our method outperforms the state-of-the-art techniques and benchmarks both in robustness and efficiency.

1. Introduction

Façade parsing has received increasing attention over the past few years [1, 7, 9, 15, 16, 22, 23] with the growing popularity of urban image databases like Google Street View and Microsoft Bing Maps. The purpose of façade parsing is to discover different regions of a façade with semantic meanings. Potential applications include reconstruction of buildings, navigation systems and location-based services, which require relatively simplified yet structurally faithful model representations. The main challenge of façade parsing lies on the fact that significant variations exist between buildings of different architectural styles. Apart from that, it can also be expected that there exist external disturbances including occlusions, noise and varying illumination conditions.

Recently a lot of research has been done to tackle this problem by leveraging the regularity of façade geometry. The key observation is that most façades in real world share strong repetitions and alignments in structure. By exploiting the regular structures in architectures, it is able to effectively parse the façades with abundant periodicity and symmetries.

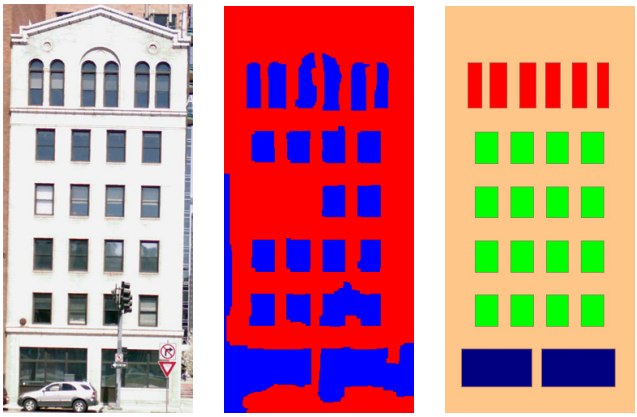


Figure 1. A façade is parsed into blocks of aligned and repetitive objects. Left: The input façade image. Middle: The initial wall/non-wall classification. Right: The parsed blocks with their repetitive objects, where a different color codes a different block.

There are two mainstreams of works addressing the problem of façade parsing: bottom-up symmetry detection and top-down parsing based on shape-grammars. In one sense, two approaches are similar in that they both impose strong prior about the regularity of structures in architectures. But they differ in many ways as well. Previous works about symmetry detection utilize local features to measure mutual similarities, where the comparison can be done in the color space [16] or feature space [13], or a combination of both [25]. In [17], Musialski et al. proposed an elegant solution to image repair based on symmetry. Other methods include generalized Hough Transform [13] and mean-shift belief propagation [18]. The major limitation of these methods is that detecting symmetries alone cannot give a model representation of façade, rendering the reconstruction difficult. Grammar-based methods resolve the difficulty by representing buildings using a set of replacement rules and a dictionary of basic shapes. Alegre and Dellarty [1] obtained a semantic description of buildings by introduc-

ing a hierarchical context free grammar. Han and Zhu [9] combined bottom-up and top-down image interpretation by using an attribute grammar. In his latest work, Shen et al. [20] proposed an adaptive partitioning method that works in both 2D images and 3D point clouds. Teboul et al. have led several works in façade parsing based on 2D shape grammars [22, 23]. Their recent method formulates façade parsing in the framework of Markov Decision Process and solves it using machine learning techniques [22]. Grammar-based parsing methods are favorable because they provide compact structured representations and offer a scalable and flexible way to do procedural modeling. In this paper we address the similar shape grammar parsing problem from a low-rank perspective with this observation: viewed as a matrix, a façade potentially has small ranks. This suggests that we could rely on the recent advancements in convex optimization theories and algorithms [2, 3, 6, 14], especially the work of Robust Principal Component Analysis by Candes et al. [6], and formulate façade parsing into a rank-one approximation problem.

Our main contribution is two-fold:

1. We provide a novel perspective to view the façade parsing problem. We seek for a balance between the simplification of representation and the fidelity of model, and propose appropriate shape grammars. Then, for the first time to our knowledge, the parsing problem is formulated into a matrix approximation problem.

2. We provide a general rank-one approximation algorithm, which can efficiently and robustly solve the parsing problem by partitioning the façade into blocks first and approximating each block with a rank-one matrix. It is worth mentioning that our algorithm not only works for façade parsing but can also serve to solve other interesting and well-discussed problems such as matrix factorization and robust clustering.

The remainder of this paper is organized as follows: section 2 introduces our motivation and problem formulation, with a description of each stage of parsing. In section 3, we describe our main algorithm to solve the rank-one matrix approximation problem. Details about block-wise parsing are discussed in section 4. Section 5 provides experimental results and compare with other approaches, demonstrating the speed and robustness of our algorithm.

2. Motivation and Overview

2.1. Motivation

By definition, single-view façade parsing aims at solving the following problem: given a façade image and a set of rules as input, how to optimally segment the image based on rules and annotate separate regions with distinctive labels.

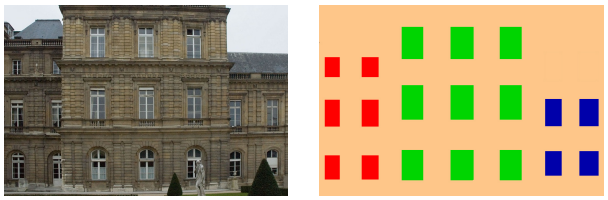


Figure 2. A façade can be partitioned into multiple blocks. Left: input façade image. Right: the image is vertically partitioned into three blocks from left to right. Each block has rank one if viewed as a 0-1 matrix.

In this paper we mainly focus on two types of labels: wall and non-wall. We notice that in previous works of Teboul et al. [11, 22], many labels such as window, roof, door, etc. are considered. We show that our simplification of considering wall and non-wall only facilitates an intuitive 0-1 matrix representation, and can be handled in the framework of rank-one approximation.

The binary split grammar [10] iteratively splits a façade into shapes with labels. In Teboul's work, it enforces shape alignment between different rows. This is reasonable in the case of façade, where windows are usually aligned; also this assumption reduces the number of states and can accelerate the optimization procedure. Likewise, we first consider the simple case of a façade, or a part of a façade, with perfect alignment between rows. Our key observation is that, considering the façade image with perfectly aligned objects as a 0-1 matrix, with 0 representing wall and 1 representing non-wall, the rank of the matrix should exactly be one, assuming that all shapes are rectangles. This observation fills in the gap between façade parsing and rank-one approximation. Therefore we can develop algorithms that can handle the general matrix approximation problem, which is then used to solve the challenge of façade parsing.

However, it is rarely sufficient to approximate real world façade with a single rank-one matrix. Defining a block as a region with repetitive and aligned element layout, a façade should have multiple blocks each containing roof, windows, doors, etc. Therefore we assume that every façade can be partitioned into blocks, and every block can be approximated by a rank-one matrix, as far as only two labels of wall and non-wall are concerned. Based on this, the façade parsing problem becomes a rank-one block-wise approximation problem. This problem seems trivial in the first place: an intuitive way is to partition each pixel into a block! However we aim at detecting repetitive patterns, and the repetitive, aligned objects should be grouped into one block (Fig. 2).

2.2. Overview

Our method contains three stages, described as follows:
Façade rectification and low-rank extraction: We use algorithm proposed in [24] to globally rectify the input façade

(a) (b) (c) (d)

Figure 3. Three stages of our parsing procedure. (a) The input façade image. (b) The rectified façade with the low-rank part, which removes the traffic light and regularizes the shapes of the first rows of windows. (c) The initial classification gives two missing windows due to different window color. (d) The final parsing result displays three blocks with aligned patterns. Two missing windows in (c) are recovered by approximating the middle block with a rank-one matrix.

image and extract the low-rank texture that recovers the regularity of the façade. The output contains a transformation and a low-rank texture. In the low-rank texture irregular objects such as cables and street signs are removed from the input.

Wall/non-wall classification: This step utilizes low-level color information and uses randomized forest as classifier to generate an initial classification for the input façade image. The output returns a pixel-wise classification of two labels: wall and non-wall.

Rank-one approximation and block-wise partition : In this step we seek to optimally partition the façade into blocks, and approximate each block with a rank-one matrix. The block-wise partition should best match with the classification generated at stage 2. Needless to say, to correctly parse the façade we highly rely on a good initial classification. The condition for a "good" classification will be discussed in detail in later sections.

The procedure is illustrated in Fig. 3. In next sections we first provide the nuts and bolts of the algorithm for rank-one matrix approximation, followed by the derived algorithm for block-wise partition.

3. Rank-One Matrix Approximation

The main algorithm developed in this paper is to compute a general rank-one matrix approximation closest to a given matrix in the sense of l^0-norm. We first examine the property of block errors and come up with the l^0-norm formulation, and then propose a fast optimization algorithm based on Augmented Lagrangian Multipliers.

3.1. Block Factorization as Rank-One Matrix Approximation

By assumption, a façade is composed of multiple blocks. The 0-1 classification matrix of each block has rank-one property. However, due to variations of pixel characteristics

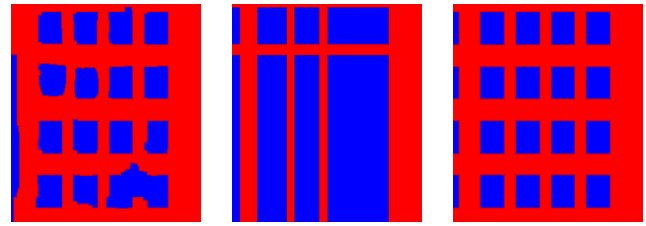


Figure 4. Comparison between the l^0-norm and l^2-norm minimization. Left: Input matrix. Middle: Rank-one approximation with PCA whose error is minimum in l^2-norm. It is rescaled to 0-1 matrix and the errors are spread. Right: rank-one approximation whose error is minimum in l^0-norm.

or occlusions, the classification results are usually corrupted with inaccurate boundaries, errors, or missing patterns. We assume that for an initial classification, only a small fraction of entries are corrupted, i.e. the classification errors are sparse. Thus our goal is to seek a rank-one 0-1 matrix that optimally approximates the given 0-1 matrix, with as few different entries between the input and output as possible.

The problem can be formulated as follows: given an input matrix M, solve the rank-one approximation:

$$\min_{M^0,E} \parallel E \parallel_0 \text{ subject to } M = M^0 + E;\ \text{rank}(M^0) = 1, \tag{1}$$

where $\parallel \cdot \parallel_0$ is the l^0-norm and denotes the number of non-zero entries. The recovered M^0 is the rank-one approximation of M, and E is the error matrix, which is optimized to be sparse. Note that M^0 is not necessarily a 0-1 matrix. But since its rank is one we can round up the non-zero entries to be one and meanwhile keep its rank unchanged.

The l^0-norm versus the l^2-norm. Principle Component Analysis (PCA) are used to seek the best low-rank representation of the given data matrix with an error matrix of minimum l^2-norm. It enjoys a number of optimality properties when the data are only mildly corrupted by small noise of Gaussian distribution [6]. However in the case of classification error, the distribution is arbitrary and the magnitude is equal to be one everywhere, so that PCA tends to spread the error and therefore fails to give a sparse response (Fig. 4).

The rank-one approximation versus the robust PCA. Recent studies in robust PCA [6, 24] suggest that convex optimization is effective and efficient for exact recovery of corrupted low-rank matrices. We have similar problem formulation and solver algorithm. However, we predetermine the expected rank rather than automatically find the minimal matrix rank. The advantage is that it is more useful to find specific patterns when we have a prior knowledge, such as façade blocks with aligned objects.

3.2. Fast Algorithm with Augmented Lagrange Multiplier

The l^0-norm is generally NP-hard, non-convex and extremely difficult to optimize [2]. Candes et al. [6] suggested that for *almost all* low-rank and sparse matrix pairs (M^0, E), under fairly broad conditions, the l^0-norm can be replaced by its convex surrogate l^1-norm, and solving the tractable problem involving l^1-norm simultaneously solves problem (1). This approach is widely adopted as the solution to l^0-norm optimization. Applying the technique here, the objective function becomes:

$$\min_{M^0,E} \parallel E \parallel_1 \text{ subject to } M = M^0 + E;\ \text{rank}(M^0) = 1, \tag{2}$$

and the l^1-norm $\parallel \cdot \parallel_1$ denotes the sum of the absolute values of matrix entries. Typical solutions of equation (2) include Iterative Thresholding [6] and Accelerated Proximal Gradient [19], which replace the equality constraints with a penalty function and try to iteratively solve an unconstrained objective function. Lin et al. [14] showed that the Augmented Lagrangian multiplier (ALM) method can be applied to effectively solve robust PCA, and outperforms other methods. Our formulation is similar to theirs. Therefore we can derive similar ALM method to solve (2).

For the ALM approach our augmented Lagrangian equation is given as:

$$L(M^0, E, Y, \mu) = \parallel E \parallel_1 + < Y, P > + \frac{\mu}{2} \parallel P \parallel_F^2, \tag{3}$$

where $P = M - M^0 - E$. Y is a matrix of Lagrangian multipliers, and $\mu > 0$ denotes the penalty for infeasible points. $\parallel \cdot \parallel_F$ denotes the Frobenius norm. Instead of replacing the rank-one constraint and put it into the unconstrained penalty function, we choose to keep it as an equality constraint. Keeping it as a standalone constraint reduces parameter tuning and simplifies the algorithm.

The basic ALM iteration scheme proposed in Bertsekas [4] switches between minimizing the augmented Lagrangian function and updating Lagrange multiplier, and is given by:

$$(M^0_{k+1}, E_{k+1}) \leftarrow \arg\min_{M^0,E} \{L(M^0, E, Y_k, \mu_k); \text{rank}(M^0) = 1\}, \tag{4}$$

$$\mu_{k+1} \leftarrow \rho\mu_k; Y_{k+1} \leftarrow Y_k + \mu_k(M - M^0_{k+1} - E_{k+1}). \tag{5}$$

By iteratively updating M^0, E to minimize the Lagrangian function while preserve the rank of M^0 to be one till converge, it automatically leads to the optimal solution of (2). In general, it is computationally expensive to minimize over M^0 and E simultaneously. We then adopt a common strategy to solve it *approximately* by *alternating minimizing* M^0, E one at a time:

$$M^0_{k+1} \leftarrow \arg\min_{M^0} \{L(M^0, E, Y, \mu); \text{rank}(M^0) = 1\}, \tag{6}$$

Algorithm 1 Rank-One Matrix Approximation via ALM

Input: Observation matrix $M \in R^{m\times n}$.
//Normalize M and initialize parameters:
$M = M/ \parallel M \parallel_F; Y_0 = M/J(M); E_0 = 0; \mu_0 > 0; \rho > 1; k = 0;$
while not converged **do**
  //Update M^0 and E:
  $M^0_{k+1} = \delta_1(M - E_k + \mu_k^{-1} Y_k);$
  $E_{k+1} = S_{\mu_k^{-1}}[M - M^0_{k+1} + \mu_k^{-1} Y_k];$
  //Update Y and μ:
  $Y_{k+1} = Y_k + \mu_k(M - M^0_{k+1} - E_{k+1});$
  $\mu_{k+1} = \rho\mu_k;$
  $k \leftarrow k + 1;$
end while
Output: M^0_K, E_k.

$$E_{k+1} \leftarrow \arg\min_{E} L(M^0, E, Y, \mu). \tag{7}$$

We now focus on how to efficiently solve (6) and (7). For (6), M^0_{k+1} can be obtained by computing M' to minimize L first, and then use SVD to compute the rank-one M^0_{k+1} with least-square error to M'. Let $\delta_k[\cdot]$ be the rank-k *shrinkage* operator: $\delta_k(M) = UV_kW'$, where $M = UVW'$ is the singular value decomposition of M, and V_k is the diagonal matrix of k largest singular values of M, by keeping the k largest diagonal entries of V. The updating scheme for M^0 is then given as:

$$M^0_{k+1} = \delta_1(M - E_k + \mu_k^{-1} Y_k). \tag{8}$$

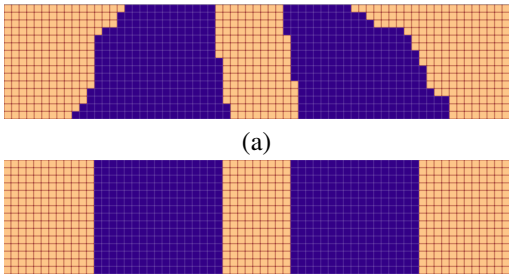
(7) can be solved accordingly. For convenience, define $S_t[\cdot]$ to be the soft thresholding operator: $S_t(x) = sign(x) \cdot \max\{\mid x \mid -t, 0\}$, where $t \geq 0$. The operator acts element-wise when applied to matrices. Therefore (7) can be solved as:

$$E_{k+1} = S_{\mu_k^{-1}}[M - M^0_{k+1} + \mu_k^{-1} Y_k]. \tag{9}$$

We summarize the ALM approach of solving (2) in Algorithm 1.

Parameter Initialization. As suggested by [14], the best initial Y_0 is $M/J(M)$, where $J(Y) = \max(\parallel Y \parallel_2, \lambda^{-1} \parallel Y \parallel_\infty)$. Major computation overhead comes from singular value decomposition. In each iteration SVD is computed twice and the total round of iteration is related to the converge speed of μ_k, which is further determined by ρ. In practice, we set ρ to 1.25 as an optimal option in experiments.

Robustness and efficiency. The algorithm is robust in the sense that it almost always returns the optimal rank-one approximation with the matrix of sparsest error,



(a)

(b)

Figure 5. The 1D rank-one approximation: (a) The input 0-1 matrix. (b) The optimal rank-one approximation with fewest distinct entries with input.

as proved by Candes et al. [6]. Such optimization is usually costly to be computed by conventional approaches. To illustrate the case, consider first how we would usually find a rank-one approximation for a single row: for each column if more than half of the pixels are 1's, we set the column to identical 1's; otherwise we set them to all 0's (Fig. 5). The 2D case is similar to 1D in concept but requires optimizing in both vertical and horizontal directions. It is NP hard in essence such that exhaustive enumeration of all possible combinations takes exponential time. Adapted methods are studied in [22]: dynamic programming would still take days on a standard machine, while random algorithms like reinforcement learning have better time performance but do not guarantee to converge to the global minimum. On the contrary, our algorithm is definite and will return the global optimal result for a large range of input. Fig. 6 shows that our algorithm works as expected in the presence of very severe missing patterns and corruptions.

We tested the running time of our implementation in MATLAB on a PC with a 2.7Ghz processor. Both the time it takes for one singular value decomposition and converge speed of iteration vary on different inputs, due to different matrix size and to what extend the matrix is corrupted. Fig. 7 demonstrates the average running time for matrices from size 50×50 to 1000×1000, from which we observe that the running time increases exponentially as the matrix size grows. It also works relatively fast for matrices below size 250×250, with average running time below 1.5 seconds.

4. Façade Block-wise Partition

We apply our rank-one matrix approximation algorithm to block-wise partition a façade, and approximate each block with a rank-one matrix. The procedure contains two stage: the pre-processing stage is rectification and classification, in which we rectify the input image, extract the low-rank part and do the pixel-wise wall/non-wall classification; the second stage is the main stage: we iteratively partition

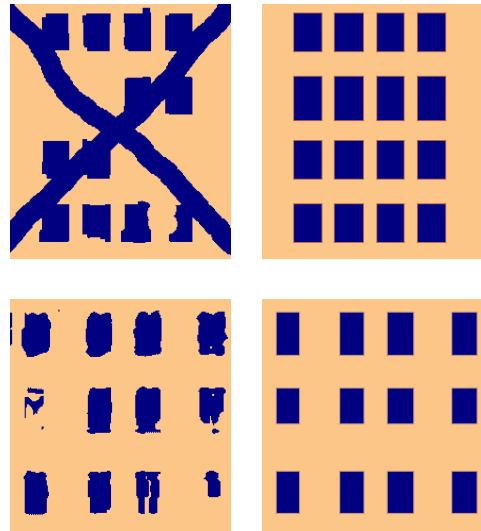


Figure 6. Illustration of robustness. Top row: an artificial test case where a cross is overlayed (left) and the recovered underlying pattern. Bottom row: a severe case with missing patterns in our experiment (left) and the underlying pattern recovered (right).

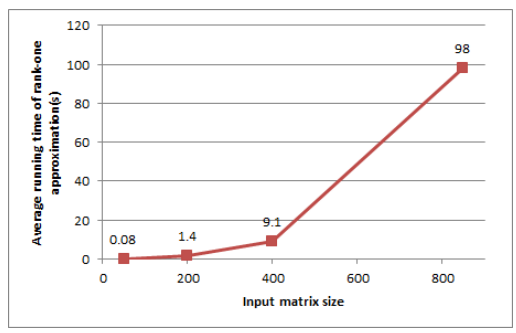


Figure 7. Relation between the running time of the algorithm and the size of the matrix. We can infer that the running time increases exponentially as the matrix size grows.

the classification into rank-one blocks.

4.1. Pre-processing

Façade rectification and low-rank extraction. Given a façade image I, we rectify I and extract the low-rank part by solving the following optimization function proposed by Zhang et al. [24]:

$$\min_{I^0,E,\tau} \text{rank}(I^0)+\lambda \parallel E \parallel_0 \text{ subject to } I\circ\tau = I^0+E, \quad (10)$$

where τ is the transformation and I^0 is the low-rank part. λ here accounts for the weight of noise, and the balance between the low-rank part and the noise part. It determines how regular the recovered low-rank image I^0 is. Without losing the structure but still removing undesired noise, we

set λ to be 0.3 in our experiments.

Randomized forest for wall/non-wall classification. This step adopts randomized forest [5] as classifier to generate a wall/non-wall classification for a given façade. Viewed as a 0-1 matrix, the output classification is then used to set up a score function for the next step block-wise partition.

The feature vector we consider here is histogram of oriented gradients (HoG) and texton for all pixels of an image. Training database contains 140 labeled façade images. Suggested by Lepetit et al. [12], we use 10 trees to build the randomized forest, In the training phase, labeled façade images are used to learn the classifiers. In the testing phase, the feature vectors of every patch of the input façade are dropped in each tree and we end up with a pixel-wise classification.

4.2. Block-wise partition

Having obtained the wall/non-wall classification 0-1 matrix C, we seek to partition it into multiple blocks and approximate each block with a rank-one 0-1 matrix. In this way we get a new pixel-wise classification with multiple rank-one structures.

We propose two heuristic methods to partition the matrix into rank-one blocks: divisive partition and agglomerative partition. The concept of both is similar to that of clustering. Both methods do not specify the number of blocks beforehand but instead rely on some empirically threshold. Before going into the partition scheme, we define a score function for any new classification C' to measure its quality, by comparing with the learned classification C from stage 2:

$$f(C') = \frac{1}{mn}\sum_{i=1}^{m}\sum_{j=1}^{n} m_{C'}(i,j), \qquad (11)$$

$$\text{where } m_{C'}(i,j) = \begin{cases} 1 & \text{if } C'(i,j) = C(i,j); \\ 0 & \text{otherwise,} \end{cases}$$

where m and n are the row and column number of the façade image respectively. Accepting C as input, we hope to partition it into multiple blocks. Consider first the choice of placing splitting planes. Based on some weak prior knowledge of urban buildings, originally observed by Mü ller et al. [2007]: *horizontal (vertical) splitting planes should be placed where vertical (horizontal) lines are rare and horizontal (vertical) lines are dense,* possible positions of splitting planes are constrained to be placed at places with large gradients. For instance, a horizontal splitting plane should be aligned with some horizontal lines with large gradient responses in y direction. In practice, gradient responses changes continuously when moving the lines from up to down, so we select the peak locations with certain mutual distances as potential positions for horizontal splitting planes. For better robustness, gradient is considered in the low-rank domain: we use the low-rank texture I^0 acquired in step 1 to compute the gradient response.

After obtaining the potential splitting planes we are able to calculate the best set of planes that partition the matrix into rank-one blocks and have large score regarding equation (11). The divisive partition starts with the entire matrix and recursively split at the optimal position, i.e. the horizontal or vertical line that yields the partition and the corresponding block-wise approximation with largest score. The agglomerative partition splits the matrix with the whole set of potential splitting planes and recursively merge them. Both methods converge when the score is larger than some threshold.

In experiment, we adopt the divisive partition method, and set the threshold for converge to be 0.9, i.e. the new classification computed should agree with the initial classification with accuracy of more than 90% . Experiments show that 84% of our test cases have no more than 6 potential splitting planes, and can converge after 3 iterations. Time cost is reasonable due to the partitioned blocks have a much smaller size comparing with the original matrix, and so our rank-one approximation algorithm takes exponentially less time. On average, block-wise partition and approximation takes twice the time of approximating the entire matrix once. To further accelerate process, it is natural to adopt a multi-resolution approach: at low-resolution level we acquire rough block-wise partition, and then approximate each block with rank-one matrices at higher resolution.

5. Experiments

We implemented the algorithm in MATLAB and ran it on an Intel i7-2620M CPU with 4GB RAM. In total we tested 54 façade images of different architecture styles. 44 of them match well with the ground-truth classification with precision above 95%. Moreover, they are able to retrieve exact information of symmetry and repetition. Some of the representative results are displayed in Fig. 10 and Fig. 11. As mentioned in Section 2, the accuracy depends solely on the quality of the classification, so imperfect classification is likely to lead to defective results (Fig. 10).

The speed of our method outperforms the state-of-the-art approaches of façade parsing under similar framework but using different approaches. For our test cases the average time for images of size below 500×500 is 6.2s. Although the average speed to approximate a single matrix of size 500×500 is about 10 seconds, we dramatically accelerate the process using a multi-resolution strategy to partition the matrix into small blocks at lower resolution. We tested identical images provided by GraPes [21], which implements shape grammar parsing based on reinforcement learning. Not only the speed is four times faster, but we

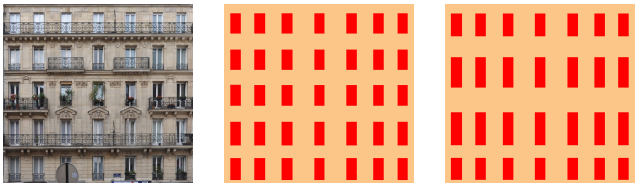


Figure 8. Comparison of parsing results with GraPes. Left: The input image. Middle: Our result. Right: Incorrect result of GraPes.

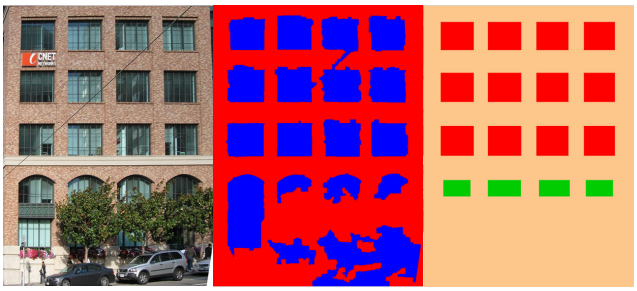


Figure 9. A failure case due to imperfect initial classification.

have a higher accuracy as GraPes does not provide consistent results every time (Fig. 8). Our optimization is also robust regarding small occlusions, under the condition that the initial classification does not lose too much of the structural information. We can also handle the case whose symmetry fails to be detected by Zhao et al. [25] (Fig. 11).

Limitations. We discuss our limitations from two aspects. First, our method is actually a global optimization approach and depends on classification result alone. It pays little attention to local features which may be more effective when there is large occlusions. Second, given varying building styles, we are not able to handle certain architectures that disagree with our assumptions of "blocks", such as those with sophisticated layouts of windows or other objects. Nevertheless, for façade elements that are in shape other than rectangle, there is no effective way to accurately segment the region.

6. Conclusions and Future Work

We have introduced a fast algorithm for rank-one matrix approximation with its application on block-wise façade parsing. As a tractable approach, the algorithm can also serve to solve other common problems. Since the quality of parsing relies on the initial classification, in the future we aim at expanding our training database to elaborate the classification. It is also possible to extend rank-one to rank-k so that the recovery of more complicated pattern is feasible.

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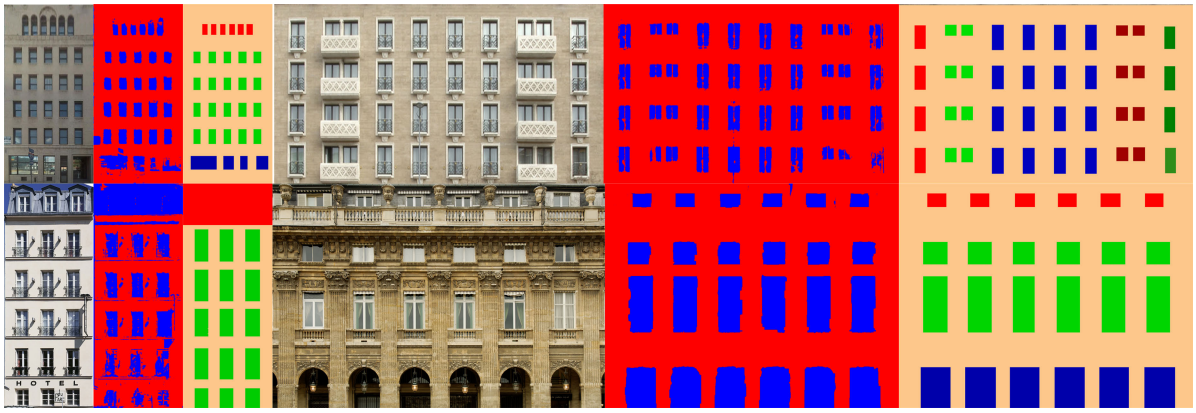

Figure 10. Parsing results of architectures of classic styles. We show the intermediate classification result as well. The bottom-right façade suffers from misalignment with the parsing result because the round-shape arches are approximated by rectangles.

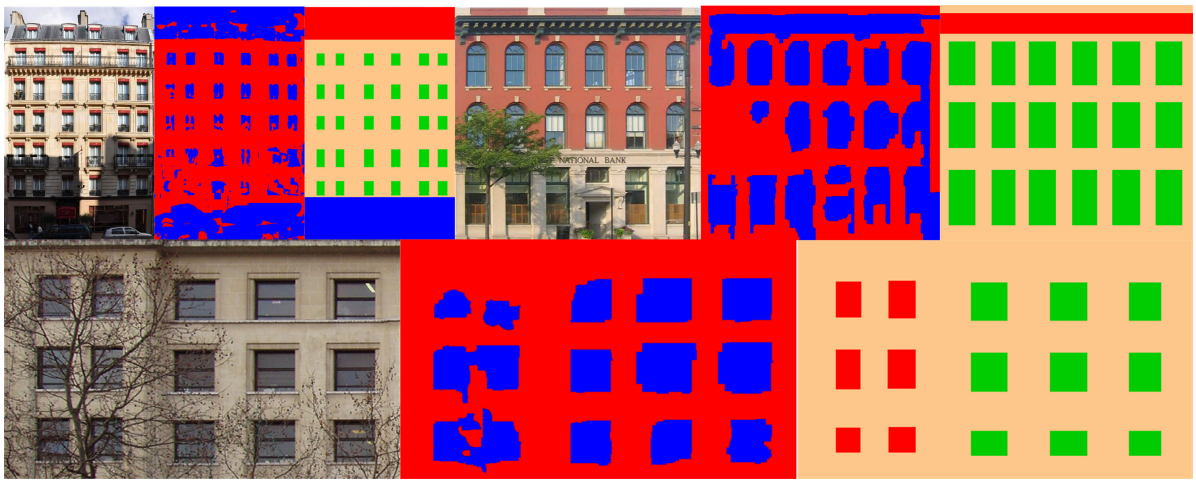

Figure 11. Parsing results of images with illumination variations and occlusions, with intermediate classification given. Top right is a failure case of the method of Zhao and Quan [25].

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